

MARKET RESEARCH BUSINESS PLAN & IMPLEMENTATION ROADMAP

AI-Based Real-Time Facial Recognition Security System

Phase 1: Retail Watchlist Alert | Phase 2: Women's Safety | Phase 3: Smart City

Document 2 of 3 | May 2026 | CONFIDENTIAL

MVP assumed: built by 2 founders. This document covers full journey from Day 0 to exit.

SECTION 1: MARKET RESEARCH

1.1 Market Sizing — TAM / SAM / SOM

■37,000 Cr TAM — India Video Surveillance 2025 Growing to ■59,000 Cr by 2030 (Mordor Intelligence)	■3,500 Cr SAM — India AI-CCTV + FRT 2025 Growing at 18-22% CAGR (NextMSC + IMARC 2025)	■150-300 Cr SOM — 5-Year Revenue Target 1-2% SAM capture by Year 5 (focused retail + Tier-2 cities)
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Market Segment	2025 Size	2030E / 2034E	CAGR	Source
India Video Surveillance (overall)	■37,000 Cr (~US\$4.4B)	■59,000 Cr by 2030	10.1%	Mordor Intelligence
India CCTV Hardware Market	US\$5.0B	US\$22.5B by 2034	17.6%	IMARC 2025
India AI CCTV Software Market	■6,900 Cr (~US\$827M)	US\$3.67B by 2030	21.8%	NextMSC 2023
India Facial Recognition Market	US\$418.5M	US\$903.6M by 2034	8.7-18.7%	IMARC / TechSci
Global AI Video Surveillance	US\$3.9B	US\$12.46B by 2030	21.3%	MarketsandMarkets Dec 2024
India Women Safety Tech (proxy: Nirbhaya Fund)	■7,712 Cr cumulative	Ongoing (MHA)	—	PIB / MHA 2024

Note: Multiple research firms report varying figures due to scope differences. Conservative Mordor figure used for financial modelling. India explicitly ranked as highest-CAGR country in global AI video surveillance by MarketsandMarkets (December 2024).

1.2 CCTV Infrastructure — India's Installed Base

India has one of the world's densest urban surveillance networks. Delhi has the highest camera density per square mile globally (1,826 cameras/sq mi per Comparitech). This installed base = addressable hardware to overlay AI software on.

City	Est. Cameras	Density	Smart City CCTV	Key Note
Delhi	313,332 public	1,826/sq mi — #1 globally	Yes	Safe City ■798 Cr, 300K-suspect FRT DB
Hyderabad	900,000+ total	79.4/1,000 people	Yes	2nd most surveilled non-Chinese city globally
Bengaluru	585,284 geo-tagged	High density	Yes	Safe City ■496.57 Cr (Honeywell) + 7,000 new cameras
Indore	251,500+	Smart City leader	Yes	84,000 Smart City cameras across 100 cities total
National Total (est.)	2-2.5M commercial/public	~20M incl. residential	100 Smart Cities	No authoritative MHA census; industry estimates only

1.3 Retail Pain Point — The Commercial Thesis

3.1% India Retail Shrink Rate World's highest (Global Retail Theft Barometer)	■23,000 Cr Annual Retail Loss Estimated against ■7.5L Cr organised retail market	0.19% India Security Spend Below global 0.31% average — structurally under-invested
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Commercial logic: India's retailers lose more than any country in the world to theft, yet spend below-average on security. A ■4,500/camera/month AI watchlist product that demonstrably reduces shrink by 10-20% delivers a 3-6 month payback for the retailer. This is a solvable ROI equation.

1.4 Government Budgets — Women's Safety & Smart Cities

Programme	Budget	Status	Relevance to Product
Nirbhaya Fund	■7,712 Cr allocated; ■5,846 Cr (76%) utilised	Active — MHA is largest implementing ministry	Phase 2 women's safety features; partner with Safe City SI contractors
MHA Safe City Project	■2,919.55 Cr across 8 metros	Ongoing; Bengaluru ■496.57 Cr awarded to Honeywell	Subcontractor AI/FRT layer opportunity; ~68% of Nirbhaya Fund
Smart Cities Mission	■1.64 lakh Cr total; 84,000 CCTVs installed	Mission closed March 2025; O&M; opportunities remain	AI-overlay SaaS on existing camera infrastructure
iDEX / ADITI 2.0	Up to ■25 Cr per challenge (ADITI); ■10 Cr (iDEX Prime)	Active — ongoing problem statements	Non-dilutive grant for Phase 1 MVP; Phase 2 defence/CAPF expansion

SECTION 2: BUSINESS MODEL

Multiple revenue models are designed for the Indian market context. They run in parallel across segments and phases.

Model	Target Segment	Pricing (Indian Context)	Revenue Type	Phase
B2B SaaS — Per Camera	Retail chains, malls, banks	■3,000–■8,000/camera/month (Staqu validated range)	Recurring ARR	Phase 1
B2B2G — Watchlist Alert API	Retail + police integration	■50,000–■2,00,000/yr per police jurisdiction API	Recurring + Usage	Phase 1-2
Edge Hardware Bundle	SMB retailers, tier-2 cities	■75,000–■1,50,000 device capex + ■2,000/month SaaS	Capex + Recurring	Phase 1
Government Sub-Contracting	Safe City SIs (L&T;, BEL, Honeywell)	■500–■1,500 per camera per year AI licensing	B2B Licence	Phase 2
SaaS — Women's Safety Module	Malls, metros, campuses	■15,000–■40,000/month per location	Recurring SaaS	Phase 2
Smart City AI-Overlay SaaS	Municipal corporations, SPVs	Tender-based; ■2–■5 Cr per city per year	Govt. Contract	Phase 3
Defence / CAPF (iDEX route)	Ministry of Defence, CISF, BSF	■10–■50 Cr government procurement	One-time + AMC	Phase 3

PRIMARY MODEL FOR YEARS 1-3: B2B SaaS per-camera/month to organised retail chains. CAC ₹3-8 lakh per logo. Target 60% gross margin. Add B2B2G police API as a free pilot Year 1, charge Year 2+.

2.1 Key Business Metrics — Indian Context

Metric	Definition	Target (Year 1)	Target (Year 3)	Benchmark
ARR	Annual Recurring Revenue	₹60-80 lakh	₹4-6 crore	StaQu ARR ₹40 Cr (Year 10); AllGoVision ₹13 Cr (Year 15)
CAC	Customer Acquisition Cost	₹4-6 lakh/logo	₹3-4 lakh (channel)	Indian B2B SaaS security: ₹3-8 lakh via direct enterprise sales
LTV	Lifetime Value (per logo)	₹20-30 lakh	₹60-90 lakh	LTV:CAC target > 3:1 (minimum); >5:1 (healthy)
ARPU	Avg Revenue per Camera/Month	₹4,000	₹4,500	₹2,000-₹8,000 published range (StaQu). Start mid-range.
Gross Margin	Revenue minus COGS	50-55%	62-68%	Indian AI SaaS: 55-70% gross margin at scale
Churn Rate	Logo churn per year	<15%	<8%	StaQu: 'zero customer churn' (company-stated, unaudited)
Payback Period	Months to recover CAC	14-18 months	10-12 months	Acceptable range for Indian B2B: 12-24 months
NRR	Net Revenue Retention	100%	110-120%	Expansion via more cameras per logo as trust builds
Burn Rate	Monthly cash consumption	₹12-18 lakh/month	₹30-45 lakh/month	Team of 6-8 in Bengaluru/Gurgaon all-in

2.2 Pricing Strategy

India is a price-sensitive enterprise market. The key is ROI articulation, not feature parity. Structure pricing as: (a) per-camera SaaS with a minimum 20-camera commitment, (b) annual contract minimum to reduce churn, (c) tiered features — Basic Watchlist Alert (₹2,500/cam/mo), Full Analytics + Women's Safety (₹5,500/cam/mo), Enterprise + Police API (₹8,000/cam/mo).

Tier	Price/Camera/Month	Features	Target Customer	Contract
Basic — WatchShield	₹2,500	Watchlist matching + alert only. Edge processing. No cloud biometric storage.	Independent stores, SMB retail	12-month min
Pro — RetailGuard	₹4,500	All Basic + repeat-offender tracking + store analytics + police alert API + DPDP audit log	Retail chains (20-200 cameras)	24-month min
Enterprise — SafeCity AI	₹7,500	All Pro + women's safety module + multi-location dashboard + SLA 99.9% + custom integrations	Large chains, malls, campuses	36-month + volume discount
Government / SI Licence	₹500-1,500/cam/yr	White-label API licence. Customer handles UI. Annual audit included.	Safe City SIs, municipal corps	Per tender

SECTION 3: IMPLEMENTATION PLAN — MVP to Scale

STARTING ASSUMPTION: 2-person founding team (1 AI/ML engineer + 1 business/product founder). No prior funding. Building from scratch.

3.1 Phase 0 — Foundation (Month 0-3, Budget: ₹8-12 lakh)

Activity	Owner	Timeline	Cost (Est.)	Output
Incorporate Pvt Ltd + DPIIT Startup	Business Founder	Week 1-2	₹15,000	GeM eligibility, iDEX eligibility, tax holiday
File iDEX SPARK / ADITI 2.0 application	Both Founders	Week 2-4	₹0	Up to ₹10 Cr non-dilutive grant if selected
Build face-detection proof-of-concept on Jetson Orin Nano	AI Founder	Week 1-6	₹50,000 hardware	Working demo: detect + match face in <2 sec
Sign 2-3 retail LOIs (pilot letters)	Business Founder	Month 1-3	₹2-3 lakh travel/demos	Proof of demand; needed for seed pitch
Engage privacy counsel (Khaitan / Ikigai Law)	Business Founder	Month 1	₹1.5-3 lakh retainer	DPDP architecture review; data flow mapping
Open founder angel round (friends/family/angels)	Both	Month 2-3	₹0 cost	₹25-50 lakh to fund Phase 0-1 operations

3.2 Phase 1 — MVP Build & First Pilots (Month 3-12, Seed: ₹4-6 crore)

Target: 3 paying retail logos, 200+ cameras, ₹60-80 lakh ARR. Raise seed round at ₹20-25 crore valuation.

Milestone	Month	Hiring Needed	Budget
Complete edge-AI MVP (face detect + match + alert pipeline)	3-5	1 embedded engineer (contract)	₹20-30 lakh
STQC certification for edge hardware device	4-7	None (outsource testing lab)	₹25-40 lakh
Deploy Pilot 1: 20-30 camera retail store cluster	5-6	1 field deployment engineer	₹8-12 lakh capex + install
Develop police-alert API (B2B2G layer)	6-8	AI Founder (part-time)	₹2-3 lakh
DPDP audit log + consent management module	6-9	1 backend developer (hire)	₹5-8 lakh
Close Seed Round (₹4-6 crore)	6-9	—	Dilution: 15-20%
Scale to 200 cameras across 3 logos	9-12	1 sales engineer hire	₹15-20 lakh deployment
First ARR checkpoint: ₹60-80 lakh	12	Team: 6-7 people	Burn: ₹12-15 lakh/month

3.3 Hiring Plan — 2 Founders to 20 People in 3 Years

Role	Hire Month	Salary (₹ lakh/yr)	Priority	Source
ML / Computer Vision Engineer	Month 4-6	18-28	CRITICAL	IIT/NIT ML grads; Bengaluru pool; LinkedIn

Embedded / Edge AI Engineer (Jetson)	Month 4-6	16-24	CRITICAL	NVIDIA ecosystem; IISc Bengaluru
Backend Developer (FastAPI/Kafka)	Month 6-8	14-22	HIGH	Bangalore / remote India
Enterprise Sales Lead (Govtech/Security)	Month 5-7	20-32 + commission	CRITICAL	Ex-Staqu/Hikvision/CP Plus channel; LinkedIn
Compliance / Legal Officer	Month 8-10	14-20	HIGH	NLSIU Bengaluru / NLU grads; privacy specialists
Field Deployment Engineer	Month 6	8-14	MEDIUM	CCTV installation background; ITI-trained
Data Scientist (Bias Audit + Model)	Month 10-12	16-25	HIGH	IISc/IIT-Bombay ML pool
Government Relations Advisor	Month 8-10	Advisory equity 0.5-1%	HIGH	Ex-IPS/IAS with MHA / Smart Cities background

3.4 Phase 2 — Scale & Women's Safety (Month 13-30, Series A: ■25-40 crore)

Triggers: ■4 crore ARR, <10% logo churn, 1,000+ cameras deployed, STQC certified product.

Initiative	Description	Expected Revenue Impact	Timeline
Women's Safety Module Launch	Harassment detection (pose estimation), SOS trigger, stalker re-identification for malls and campuses	+■50-80 lakh ARR Year 2	Month 15-20
Tier-2 City Expansion	Target cities: Coimbatore, Nagpur, Surat, Kochi, Jaipur. Lower competition, 40-50% lower CAC	2x addressable market	Month 18-24
Safe City SI Partnership	Approach L&T; SmartWorld, BEL, NEC India as technology subcontractor for Nirbhaya Fund projects	■1-5 Cr/city contract	Month 16-24
Bias Audit Publication	First-party bias study on Indian skin-tone dataset; publish openly. Builds credibility and legal defensibility.	Brand / compliance moat	Month 12-15
Series A Close	■25-40 crore at ■100-150 crore valuation	18-month runway	Month 18-24

3.5 Phase 3 — Smart City & Defence (Month 30-60)

Targets: Smart City AI-overlay contracts, iDEX defence procurement, international expansion to Middle East and Southeast Asia (Indian brands have post-Chinese-ban advantage). Revenue goal: ■50-100 crore ARR by Year 5.

SECTION 4: FINANCIAL MODEL & CAPITAL PLAN

4.1 Five-Year Revenue Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Paying Cameras	200	800	2,500	6,000	12,000

ARPU (■/cam/month)	3,800	4,200	4,500	4,800	5,000
SaaS ARR (■ lakh)	91	403	1,350	3,456	7,200
Government / SI Revenue (■ lakh)	0	0	200	800	2,500
Hardware Sales (■ lakh)	30	60	100	150	200
TOTAL REVENUE (■ lakh)	121	463	1,650	4,406	9,900
Gross Margin %	52%	58%	64%	67%	70%
EBITDA (■ lakh)	-180	-120	80	660	2,200
Burn Rate (■ lakh/mo)	12-15	25-35	40-55	60-80	80-100

Model assumptions: 18-month sales cycle for govt; 5% hardware margin; ARPU grows with product maturity; EBITDA turns positive Year 3 with 2,500+ cameras.

4.2 Capital Structure — Working Capital & Fixed Capital

Capital Item	Type	Year 1	Year 3	Notes
Edge Hardware (Jetson + cameras per deployment)	Fixed / COGS	■30-50 lakh	■1.5-2 Cr	Partly recoverable via hardware sale or lease to customer
R&D; (model training, compute, dev)	Working Capital	■15-20 lakh	■60-80 lakh	GPU cloud (AWS/GCP) + on-prem GPU server ■8-12 lakh
STQC / BIS Certification	Fixed (one-time)	■25-40 lakh	■5 lakh/yr maintenance	Mandatory for all connected devices from April 2026
Personnel (salaries + ESOPs)	Working Capital	■60-90 lakh	■2.5-3.5 Cr	6-8 people Year 1; 20+ people Year 3
Sales & Marketing (demos, events, travel)	Working Capital	■15-25 lakh	■60-80 lakh	Channel partner commissions 10-15% of first-year ACV
Legal & Compliance (DPDP, privacy counsel)	Fixed + Working	■10-15 lakh	■20-30 lakh	Critical — cannot be deferred
Office + Infrastructure	Fixed	■5-8 lakh	■15-20 lakh	Co-working Year 1; leased office Year 2+
TOTAL CAPITAL REQUIREMENT	—	■1.6-2.5 Cr	■5-7 Cr	Covered by seed + iDEX grant + revenue

4.3 Funding Roadmap

Round	Amount	Timing	Dilution	Key Investors to Target	Use of Funds
Angel / Friends & Family	■25-50 lakh	Month 0-2	5-8%	Founders' networks; angel networks (ah! Ventures, LetsVenture)	POC hardware, legal, LOIs
iDEX SPARK (Non-dilutive)	■1.5-10 Cr	Month 2-6	0%	iDEX / DDPMoD — defence challenge application	MVP build, STQC cert
Seed Round	■4-6 Cr	Month 6-9	15-20%	Yali Capital, Pravega Ventures, WaterBridge, Blume, Endiya, IvyCap	Scale pilots, 3-4 hires, marketing

ADITI 2.0 (Non-dilutive)	Up to ₹25 Cr	Month 12-18	0%	MoD ADITI challenge (deep-tech public safety)	Phase 2 product, defence expansion
Series A	₹25-40 Cr	Month 18-24	20-25%	Elevation Capital, Kalaari, Sequoia India, Florintree, IAN Alpha Fund	Women's safety module, Tier-2 expansion, SI partnerships

SECTION 5: POTENTIAL INVESTORS

Investor	Type	Thesis Fit	Portfolio Ref.	Contact Approach
Yali Capital (Bangalore)	Deep Tech VC	HIGH — explicit 'Aerospace & Surveillance' thesis; pre-seed to Series A	yali.vc	Cold intro via IISc / IIT network; direct pitch
Pravega Ventures	Deep Tech VC	HIGH — backed Tonbo Imaging (₹175 Cr Series D 2025); defence + AI imaging focus	Tonbo, others	Tonbo Imaging team connection; LinkedIn founder outreach
WaterBridge Ventures	Sector VC	HIGH — Manish Kheterpal publicly bullish on defence/surveillance deep tech	—	Direct email; IIM Ahmedabad network
Blume Ventures	Early Stage VC	MEDIUM — backed deep tech; India-focused; ₹1,000 Cr fund III	Multiple AI SaaS	VC Track programme application; warm intro via portfolio
Endiya Partners	Early Stage VC	HIGH — enterprise/industrial-tech specialist; AI-first portfolio	endiya.com	Apply via endiya.com/apply; NASSCOM connection
IvyCap Ventures	Deep Tech VC	MEDIUM-HIGH — deep tech focus; India-domiciled LPs	Multiple	IIT alumni network; warm intro preferred
IAN Alpha Fund	Angel + VC	HIGH — led BigEndian Semi \$6M pre-Series A for surveillance SoC (similar domain)	BigEndian Semi	Indian Angel Network events; warm intro via portfolio founders
Florintree Advisors	Late Stage VC	MEDIUM — led Tonbo Series D, ideaForge Series B; surveillance/defence DNA	Tonbo, ideaForge	Relevant for Series A+; Tonbo referral path
iDEX / DDPMoD	Govt. Non-Dilutive	HIGH — SPARK (₹1.5-10 Cr), ADITI 2.0 (₹25 Cr) for defence/security AI	250+ startups funded	Apply at idex.gov.in; DISC problem statement application
MeitY Startup Hub / TiE	Govt. Grant	MEDIUM — AI/deep tech grants; ₹25 lakh-₹1 Cr range	—	meitystartuphub.gov.in; TiE Delhi/Bangalore chapters
SIDBI Startup Fund	Debt + Quasi-Equity	MEDIUM — MSME/startup loans ₹10-50 Cr at concessional rates	—	sidbi.in/startup; Startup India portal referral

SECTION 6: GROWTH STRATEGY & MARKET FIT

6.1 Product-Market Fit Assessment

PMF Signal	Status	Evidence
Willingness to Pay (Retail)	VALIDATED	Staqu charges █2,000-8,000/cam/month to Indian retailers with zero churn (company stated). India's world-highest shrink rate = clear ROI argument.
Police Demand (B2B2G model)	VALIDATED (indirect)	8 state police forces using Staqu; NCRB AFRS tender shows demand; but procurement broken. B2B2G sidesteps this.
Women's Safety Funding	BUDGET EXISTS	█7,712 Cr Nirbhaya Fund (76% utilised). Demand confirmed by government spend. Access requires SI partnership.
Privacy-First Positioning	UNCONTESTED	No incumbent has built DPDP-by-design edge architecture. STQC ban on Chinese vendors creates hardware vacuum. Window is 12-18 months.
Customer Willingness to Pay (Police-direct)	NOT VIABLE (Y1-3)	NCRB AFRS tender stalled 5+ years. Qualification criteria block startups. Not a near-term revenue source.

6.2 Growth Levers

• Channel Partnership

Partner with established CCTV installers/SIs (those who used to sell Hikvision/Dahua and are now searching for AI-software alternatives). This gives instant access to retail relationships. Commission: 12-15% of Year 1 ACV.

• Retail Chain HQ Sales

One enterprise sale to a 500-store retail chain HQ = █1.5-2.5 Cr ARR in one deal. Focus BD energy on 10-15 national retail chains with structured security procurement.

• Government Reference Customer

One confirmed police pilot (even free) creates enormous credibility for all other sales. Prioritise getting one state police force onto a free 90-day pilot in Year 1.

• Bias Audit as Marketing

Publishing a rigorous first-party bias study on Indian faces makes you the most credible FRT vendor in a market where no competitor has done this. Use it in every sales conversation.

• STQC Certification as Barrier

Being among the first AI-camera software vendors with STQC certification creates a real, regulatory moat — uncertified competitors literally cannot sell in India post April 2026.

6.3 Exit Strategy (5-7 Year Horizon)

Path	Acquirer Profile	Estimated Valuation Multiple	Timeline
Strategic Acquisition	CP Plus / Aditya Infotech (post-IPO; needs AI software layer); NEC India (wants India FRT presence); L&T; SmartWorld (needs AI subcontractor); Honeywell Building Technologies	8-12x ARR	Year 5-7
IPO (NSE Emerge / BSE SME first)	Public market; comparable: ideaForge (IPO 2023), Aditya Infotech (IPO 2025)	15-25x ARR	Year 6-8
International PE/VC Secondary	Warburg, General Atlantic, Temasek (Singapore sovereign); PE buyout of profitable govtech SaaS at scale	10-15x ARR	Year 5+

Sources: Mordor Intelligence, IMARC, MarketsandMarkets, NextMSC, TechSci Research; PIB/MHA/iDEX government publications; Tracxn, Inc42, Getlatka company financials; Global Retail Theft Barometer; Comparitech surveillance data; NIST FRVT; company websites (Staqu, Videonetics, Innefu, Tonbo). May 2026.